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Authordraft - Service Design Methods for the Design of Smart Surfaces

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Abstract. Smart environments, integrating different sensors and other technology solutions, are one of the key direction in which everyday spaces in buildings are expected to develop in the future. One aspect of smart environments includes large interactive surface areas, which are embedded to walls, tables or other furniture. This chapter introduces how service design methods have been applied in our research on smart surfaces, and discusses different design methods and their strengths in this context. Specifically, we present three service design cases, interactive wallpaper, a smart building infrastructure and a smart cafe table, and how the related use cases have been studied with service design methods. These three design cases are positioned at different maturity levels of the product development process, the interactive wallpaper and smart building representing early design concepts, and the interactive table a functional prototype.

Keywords: Smart Surfaces · Prototyping · Electrochromic displays · Service Design · Human-Computer Interaction

1 Introduction

Our everyday environments are becoming increasingly embedded with technology. This includes mobile devices, internet, and social media, which have taken a significant role in our daily routines and interactions, but also includes intelligence embedded in the physical environment around us. Smart environments including sensors, displays, and connected artifacts are an integral part of today's building planning. This includes homes, offices, public spaces, and smart building infrastructures. Public displays are invading both indoor and outdoor urban spaces (Kostakos & Ojala, 2013), and span from the resolution of one pixel to building-sized media walls and high-resolution installations. In the emerging domain of smart surfaces, rather than being add-on elements, displays and user interaction mechanisms are seamlessly integrated into our everyday environments. As well as conventional touch interaction, such smart surfaces can utilize a variety of sensors sensors for interaction, e.g. to react to the proximity or gestures of the user.

In this chapter, we focus on smart surfaces, as interactive surfaces that can include both input and output elements. In particular, we explore electrochromic materials as the enabling technology, and discuss its feasibility for end-user applications. Electrochromic materials change their optical properties, such as color, when an electrical voltage is applied. When formulated as an electrochromic ink and applied to a surface, an electrochromic display can be created which follows the contours of the applied surface. Our research has been conducted in close collaboration with other research disciplines, e.g. chemistry and industry partners, and with end-user applications and product design in mind as the final research goal. The research is part of the DecoChrom project (DecoChrom, 2022), which addresses different aspects of electrochromic display development, including the sustainability of the technology. This chapter focuses on service design, and introduces how service design methods have been applied in our research on smart surfaces and discusses different service design methods and their strengths in this context.

During the recent decade, service design has been adopted as a new approach for consumer-centric service product creation (Miettinen, 2017). Building essentially on the tradition of industrial design and user centered design, it applies methods from design and user research, considering the holistic user experience with the service. Service design also takes into account different stakeholders, and the design process considers the interplay between them, beyond designing for an individual user. This means taking into account the viewpoints of the consumer, service provider, and those interfacing to the consumer in the service interface. For instance, when designing a mobile sport coaching app for young athletes, in addition to the teenage mobile app user, the service design process takes into account stakeholders such as the sports coach, medical support, and the service platform handling the data (Häkkilä et al., 2016). As a consequence, the design process needs to consider a more complex contextual framework of users and a wider set of other stakeholders than traditionally considered in human-computer interaction design cases (Lee, Yap, & Roto, 2022).

Fundamentally, interactive systems designed for today's world are integrated services, rather than individual products (Forlizzi, 2018). Thus, it is important to consider service design aspects already in the early phases of the product design process. In the following, we present how service design thinking and methods have been applied in our research on smart surfaces. As design cases, we present the concept of an interactive wallpaper (Brun, Lehtinen, & Häkkilä, 2021), and two cases demonstrating the use of interactive surfaces to support the smart social distancing required during the Covid-19 pandemic, i.e. interactive surfaces in a smart building and a smart cafe table (Colley, Gröhn, & Häkkilä, 2021). In each of the three design cases, design probes with different levels of physicality were used (Figure 1).

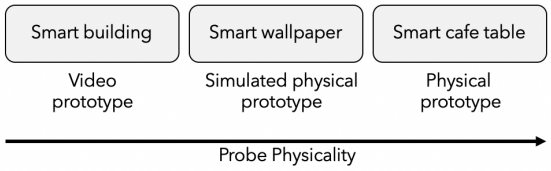


Fig. 1. The 3 presented case studies demonstrate the use of prototypes with different levels of physicality as probes in the service design process.

2 Electrochromic Smart Surfaces

Mark Weiser presented a vision of 21st century computers as computers that will "find their way invisibly into people's lives" (Weiser, 1999). This means that we will no longer carry around devices such as smartphones, but we will be surrounded by an environment containing smart surfaces and objects that we can interact with. However, current technology is not suited for this, and new material developments are needed to realize this vision (Abowd, 2020). In the future, the very surfaces that we use to construct products or buildings will become inherently intelligent. In this chapter we focus on one specific technology that has the potential to form a building block of smart surfaces - electrochromic materials.

2.1 Electrochromic Technology

Electrochromism is the capability of some materials to change their optical properties through chemical oxidation or reduction when an electric current is applied to them (Granqvist, 2015). This means that electrochromic (EC) materials can change their optical absorption characteristics or color when an electrical voltage is applied. This behavior is similar to the more well known thermochromic materials, that change color based on temperature. A variety of different electrochromic materials have been explored that can switch between different color combinations and intensities. One of the most commonly used is PEDOT:PSS (the chemical name poly(3,4-ethylenedioxythiophene) polystyrene sulfonate), which can change its color from nearly transparent to a dark blue in less than 2 seconds when a small voltage is applied (Jensen, Colley, Häkkilä, Pinheiro, & Löchtefeld, 2019).

EC materials have a variety of different properties that are extremely favorable when designing large interactive smart surfaces. First of all, EC displays operate at low voltages (usually between 1-3 Volts) and do not require complex control electronics, such as those needed for E-Ink or OLED displays. Furthermore, these displays are bi-stable, meaning once the change in color has been performed, the display will remain in this state for several hours without the need for additional power. EC materials are not light emitting, making them suitable for environments such as the bedroom, where artificial light may have negative impact on human sleep patterns (Czeisler, 2013). Finally, EC displays are very

thin (around 3 mm), flexible and can be produced in in any shape (Jensen, Colley, Häkkilä, et al., 2019), which enables their integration into a variety of different products as user interfaces (Colley, Genç, et al., 2021) or soft robots (Löchtefeld & Milthers, 2021), and are specifically suited for interactive smart surfaces.

2.2 Design with Electrochromics

Electrochromics also have several advantages when it comes to designing and prototyping EC displays. Prototyping can be done using off-the shelf printing techniques such as screen printing or a slightly modified ink-jet printer (Jensen, Colley, Häkkilä, et al., 2019). The process of assembling the displays is easy and does not require any special requirements such as a clean room, making it relatively easy for designers to test their ideas in practice (Jensen, Craft, Löchtefeld, & Bjørn, 2022).

Given that the display itself can be transparent, a wide variety of application scenarios are possible. The simplest being to add an EC display on top of a standard printed product. For example, in (Jensen et al., 2020), EC displays are integrated onto the printed graphics of a classic board game to create new interactive experiences. The fact that EC displays are flexible and can be produced in every shape has been also enabled their integration into clothing and wearable devices such as shoes (Jensen, Colley, & Löchtefeld, 2019), jackets (Genç, Kantola, & Häkkilä, 2020) and face masks (Genç, Colley, Löchtefeld, & Häkkilä, 2020). These examples demonstrate the potential of the electrochromic technology and the ways in which designers can leverage it.

3 Service Design as an Approach

Service design has its roots in design research. Whereas the field of product design has addressed physical artifacts, service design has shifted the focus beyond the traditional design scope. Instead of focusing on tangible design artifacts, service design applies the design methods and design thinking in creating and improving services. It aims to create and improve services around both immaterial and material products (Miettinen, 2012). In the area of human computer interaction (HCI), service design has commonly been interpreted as designing digital instead of physical products, and with a focus on the consumer journey and system level design (Yap, Lee, & Roto, 2021). Even though digital applications are today commonplace in all sectors of life, service design can also address cases which do not have digital components, and often it concerns hybrid products with both physical and digital elements.

Service design applies methods based on those from the tradition of user centric design. In essence, a user centric design approach means applying a design process, which keeps the user in its core focus, and involves target users through different study methods throughout the whole design cycle. From a historical perspective, design principles for creating interactive systems first focused on human factors and ergonomics, and after that, started addressing the wider

concept of usability, i.e. the efficiency, effectiveness and user satisfaction, with the system. It was however realized that these aspects are not enough when investigating users' interaction with the system and motivations behind the use. The development in the field thus proceeded to consider factors beyond usability, and to design for a holistic user experience which includes both utilitarian and hedonic aspects (Hassenzahl & Tractinsky, 2006). User experience design emphasizes aspects that do not focus only to the instrumental value of the interactive artifact, but include aspects such as engagement, emotion and affect when interacting with the system (Law, Roto, Vermeeren, Kort, & Hassenzahl, 2008). Service design considers yet a larger framework, as it broadens the focus from an individual user and takes into account also other stakeholders (Forlizzi, 2018). Lee et al. (2022) have presented four dimensions for service design from interaction level to system level, which are 1) the customer experience, 2) backstage work, 3) collaborative network, and 4) ecosystem and infrastructure. The viewpoint of the service provider is important to take into account in order to enable creating services, which result in functional and seamless experiences for the end-user. Typically there also exists other stakeholders, who have dependencies with or influence over the service. Also, it is important to remember that services do not operate in isolation, but can be seen as part of a larger ecosystem and utilize some infrastructure. It has been pointed out that HCI oriented research on service design often focuses on the interaction dimension, whereas the system level dimension includes tools and methods that are more business-oriented (Lee et al., 2022).

Service design differs from conventional interaction design also in the manner in which it considers the holistic consumer journey during which the user interacts with the system. The interaction with the system is not anymore studied as an isolated phenomenon, as is often the case with usability studies and interaction design research, but interaction is considered as part of a longer path, during which the user interacts with the service at several touchpoints. This includes not only the actual core transaction or task completion, but also the pre and post service experiences. Methodologically this means that a service design study must be designed to cover a larger sequence of events, during which the user interacts with the system. For instance, the designers can organize a study which looks at searching and finding the service, taking it into use as a first time user, and completing the actual transaction with the service.

A variety of tools and methods commonly applied in service design have been charted by Alves et al. (2013). In the analysis, gathering 164 methods and tools used in service design, the most common ones included scenarios, storyboarding, prototyping, consumer journeys, shadowing, and service blueprints (Alves & Jardim Nunes, 2013). These reflect the general approach in service design, which emphasize understanding the users and the use cases, and investigating the service as a holistic experience. Service design is also experimental and agile in its very nature, as reflected through service prototyping and scenario workflows.

To organize studies covering a significant part of the consumer journey or to concept a holistic service, different methods to simulate the service and user in-

teraction with it have become especially useful. Service prototyping is a method to where the service is simulated or set-up as a demo in a place or condition resembling the actual use context. This way, the service can be tested and its features iterated before taking it into real use. An example of a service prototyping tool is a service design laboratory space, where different use contexts can be staged with the help of back-projected screens. For example, two screens placed at a 90 degree angle can form a corner, where different use scenarios can be staged and acted out (Rontti, Miettinen, Kuure, & Lindström, 2012). Projected presentations of service spaces have been found to facilitate collaboration (Colley, Suoheimo, & Häkkilä, 2020). With a portable set of equipment, the stage can be set up at different locations, for instance at a project collaborator's or service provider's office space for a joint service design session (Häkkilä, Colley, Inget, Alhonsuo, & Rantakari, 2015). Wizard-of-Oz techniques can also be used to simulate a functional service (Meurer, Pakusch, Stevens, Randall, & Wulf, 2020). In addition, virtual reality (VR) can be utilized to investigate the service context and allow remote access to it, although this may limit the collaborative and participatory nature of the design process, and thus is not optimal in all design phases (Colley et al., 2020).

In the following, we present case studies where we have used a service design approach to investigate the potential for smart surfaces. The examples apply different methods for service prototyping, each presenting different maturity level and allowing different amount of interactivity.

4 Service Design Cases with Smart Surfaces

4.1 Case 1: Interactive Wallpaper

As the first design case adopting service design methods, we present the case of interactive wallpaper. The concept of interactive wallpaper has already been explored with the aims of providing either useful, engaging or entertaining ambient experiences (Huang & Waldvogel, 2005; Campbell, Felton, & Hoarse, 2014; Hoare, Campbell, Felton, & Betsworth, 2015). However, despite showing the potential applications of using interactive wallpapers, the previous works were mostly agnostic regarding the technical implementation, and many years or decades later they remain too far-fetched to be implemented. With this background, we addressed the topic of interactive wallpaper based on the technical possibilities of electrochromic inks (Section 2.1).

An exploratory study (Brun et al., 2021) was conducted to collect user perceptions and qualitative feedback while emulating the opportunities (e.g., low consumption with solar panels integrated into the design) and limits (bi-state monochromatic graphical elements) of this technology in five interactive wallpaper concepts, each aiming to address different ideas:

- *Timer*, in our case taking place in the bathroom, with slowly moving graphics to emphasize the required duration for hand washing.

- *Notification*, taking place in the office with the visual form of portholes notifying the user of either an upcoming meeting, the microphone status, or as a reminder for physical activity.
- *Voting*, or more broadly the idea of communal expressive walls, visually rendering the results of voting at the corners of a room.
- *Noise*, with a colored repeated pattern showing the current ambient sound level (in our case situated in a library).
- *Playful*, taking the form of a hide-and-seek game with integrated solar panel elements and a repeated pattern of animals.

The concepts were simulated in the service design laboratory, where the wallpaper concepts were visualized on two large screens. Two back-projected walls were used to display the interactive wallpaper concepts, complemented with physical furniture (tables, chairs) and tangible artifacts (keyboards, tablets, books and a sanitizer dispenser) to improve the realism of the simulation (Figure 2).



Fig. 2. The instructor setting up the additional furniture next to the two back-projected walls (left) and two participants giving feedback to the instructor during the *noise* concept scenario taking place in a simulated library (right).

In seven sessions, fourteen participants (9 women) were recruited in pairs from the university campus from students and administrative employees. For each session, the participants first gave their informed consent and were then introduced to physical electrochromic displays samples. Then, participants were invited to engage with a think-aloud protocol for each simulated concept presented in a set order as part of an imaginary short story. The narrative was such that the participant just arrived at work and had to wash their hands in the bathroom with the *timer* concept. Then, they sat in an office with the *notification* concept and transitioned to a meeting room incorporating the *voting* concept. Finally they stopped at the library with the *noise* concept (including simulated audio) before going home and engaging with the *playful* concept. One moderator gave explanations of each concept and took notes, whilst another was responsible for changing the physical structure of the settings according to each

text context. The *Wizard of Oz* method was used to simulate interactions, such as activity (e.g., washing hands) or touch detection.

Hassenzahl has defined user experience (UX) as "*a momentary, primarily evaluative feeling (good-bad) while interacting with a product or service*" (Hassenzahl, 2008). Thus, after completing the test scenarios, participants completed a questionnaire to assess their feeling toward each wallpaper concept (1-5 scale), and to collect their least and most preferred concept (including explanation of the reasons) as well as demographic data and their additional free form comments. Each session was video-recorded for subsequent analysis and lasted around 45 minutes in total. The methodology allowed us to receive preliminary feedback, both quantitative and qualitative, and their analysis provided the opportunity to highlight and further develop the two most favored concepts, *Noise (library)* and *Playful (animals)*.

4.2 Case 2: Smart Building

As the second case, we present a service design case where the user interacted with smart building infrastructure, supporting precautions for social distancing. Here, the design context was inspired by the Covid-19 pandemic. Digital technologies have been widely harnessed to fight against Covid-19 both in healthcare (Whitelaw, Mamas, Topol, & Van Spall, 2020) as well as to support society during periods of lock-down and social distancing (He, Zhang, & Li, 2021). In our design case, a smart building was considered as a service, integrating different technologies which enabled the creation of a safe zone, minimizing the infection risk for those moving through the space. With our focus on smart surfaces with integrated display elements, scenarios including safety instructions and indicators were in the main interest.

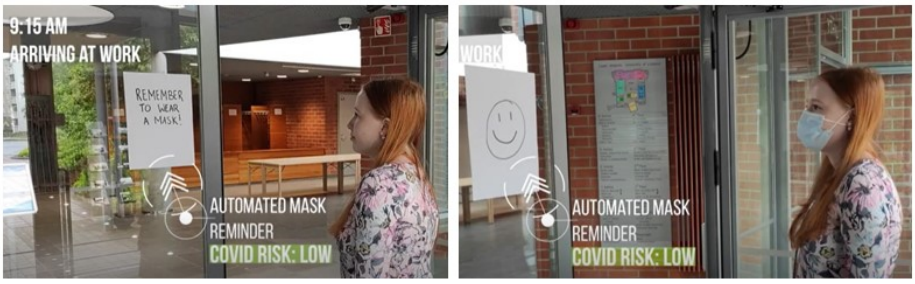


Fig. 3. Screenshots from the created consumer journey video illustrating a low-fi simulation of a smart door display that checks if a face masks is worn.

A consumer journey description was created in the form of a video, which demonstrated different touchpoints with the smart building service. The service concept included display screens which were integrated into the building. An

example of one of the concept scenarios is presented in figure 3, where the user enters the building, and the smart display at the door reminds them to wear a face mask. When the system detects the user is wearing a face mask, the display shows a smiley face icon. In the consumer journey video, the technology is illustrated using a low-fi paper prototype, with overlaid textual explanations. The video collected together several use cases for smart building technology and provided a holistic overview, which could be used to communicate the possibilities of technology to different stakeholders, such as citizens and technology companies, in an easily digestible form.

4.3 Case 3: Interactive Cafe Table

The surfaces of tables in in cafes, and public spaces such as trains and libraries provide a potential mechanism for the spread of the Covid-19 virus. To reduce this risk, the table surface should be cleaned between each user. In the *CleanLeaf* table concept, the table surface is a smart surface which visually indicates its status as either clean or dirty. In this way arriving users can be directed to clean tables and staff can directly identify which surfaces require cleaning. Two approaches to integrating indication to the table surface were explored, a) LEDs under a thin wood veneer and b) electrochromic displays integrated to the surface. With both approaches, the design target was that the display should form an integral part of the surface design, rather than being simply an added indicator.

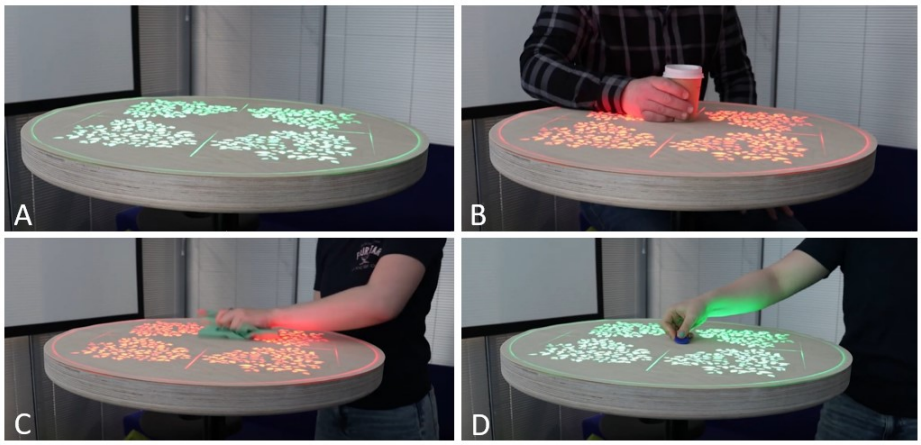


Fig. 4. CleanLeaf Table, demonstrating different patterns in a simulated version.

To evaluate the concept, functional physical prototypes of the *CleanLeaf* table were developed(Figure 4). Passive infrared (PIR) sensors integrated at the top of the table leg were used to identify if a user was static at the table for an

amount of time indicative of actual usage. When usage was detected, the surface display of the tables changed to indicate the table was dirty (red illumination or filled electrochromic leaf motifs). After cleaning the table surface, staff could reset the surface indication by placing an NFC tag in the center of the table, where and NFC reader was located under the table surface. A clean table was indicated by green illumination or empty electrochromic leaf motifs). A delay of 5 seconds was provided such that the cleaner could exit from the range of the PIR sensors after resetting the table's state.

5 Discussion

The design cases described in this chapter provide insights into how different service design methods have been used in developing service concepts for smart surfaces with integrated displays. These design cases have emphasized service prototyping, and showcase different approaches for it. In the consumer journey based case, the technology is illustrated using very simple low-fi prototypes, but the video method presents them in an authentic use context. The smart wallpaper concepts were visualized in a simulated context, which allowed a relatively immersive and easily configurable set-up in which to organize a user study. The interactive cafe table concept includes a functional, interactive prototype. Compared to two other design cases, the cafe table is at a high maturity level. It offers a plausible demonstration of the concept and provides an almost end product-like user experience, but on the other hand it demanded significantly more effort to design and implement. This makes the prototype more costly and harder to modify when design iterations are needed.

Both the interactive cafe table prototype and the smart building consumer journey video have been used for communication purposes when discussing the concepts with companies producing related products and services. Prior research has pointed out that service designers use visualizations to articulate insights, communicate insights and to maintain empathy, and they tend to use more polished visualization when communicating externally compared to internal communications (Segelström, 2012). This is also valid in our work.

Forlizzi has called for a more service-oriented approach to replace the conventional way of thinking in human computer interaction research (HCI), which has emphasized a single user and taken solely the user's perspective, rather than also considering other stakeholders (Forlizzi, 2018). Reflecting against the framework presented by Lee et al. (2022), the service design cases presented here are rather interaction design oriented, and consider less the system level design aspects. However, they already take a wider perspective to the use of smart surfaces, including several actors in the use scenarios, such as waiter and cafe customer, or following a sequence of user actions during a consumer journey. Studying the interactive wallpaper concepts on a simulated stage, where a bodystorming technique (Oulasvirta, Kurvinen, & Kankainen, 2003) was applied, emphasize the participatory design aspects that are essential in service design. In bodystorming, participants engage to the scenarios with their whole body presence,

and can act them out. In its overall approach, our research follows a similar path to that identified by Yap et al., who chart the interface between HCI and service design in general (Yap et al., 2021). System level thinking becomes more important when developing the concepts targeting the use of particular technologies, here, electrochromic displays. However, differing from most HCI research utilizing service design (Yap et al., 2021), we have focused not on digital services but on digital-physical systems. In our experience, integrating this approach has functioned well in the design process.

6 Conclusion

In this chapter we have presented three example cases of how service design methods have been applied to study and communicate concepts exploring smart surfaces. The service design cases demonstrate three different fidelity levels of prototypes, from paper prototyping presented on a video, to a simulated service scenario, and finally a functional prototype. From an industry perspective, these methods provide tools to explore and evaluate applications of technology from the user viewpoint at an early development phase. The service design approach provides a broader picture of the user experience than a single user study, and helps in orientating towards system level thinking.

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